GRADE 1 · UNPLUGGED SEQUENCING

Does the Order Work?

Read the algorithm and check the order

Name: .... Date: ....

★ **I can tell if an algorithm's steps are in an order that works.**

Check the order

Read each algorithm. Does the order make sense? Circle YES if it works, or NO if a step is in the wrong place. Then tell why. A step that needs another step must come AFTER it.

Algorithm A — draw a sun

1. draw a circle
2. add rays around it
3. colour it yellow

1 Does A work?

Circle YES or NO. Tell why the order does or does not work.

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Algorithm B — have cereal

1. eat it
2. pour the milk
3. pour the cereal in the bowl

2 Does B work?

Circle YES or NO. Tell what is wrong with the order.

Algorithm C — go down the slide

1. climb the ladder
2. slide down
3. sit at the top

3 You try — C does not work

Circle YES or NO. Then write the 3 steps of C in an order that DOES work.

Big idea

An algorithm only works if every step comes after the steps it needs. If a step is in the wrong place, the order does not work — even if all the steps are there.

TEACHER GUIDE · GRADE 1 · UNPLUGGED SEQUENCING

Does the Order Work?

Quick facilitation notes & answer key

What this worksheet teaches

Students read three plans and judge whether the order works, explaining why, then rewrite a broken one. Reading a plan and deciding if the order succeeds — and fixing it — is the heart of debugging. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: cut the steps into cards so students can physically slide them into order before writing.

How to lead it (about 20 minutes)

1. Show them (you do). Read a plan aloud and ask: "does each step have what it needs?"
2. Do one together. Plan A works; Plan B does not (you cannot eat before you pour).
3. Let them try. Students judge Plan C and rewrite it so it works.

Answer key (these are the verified correct answers)

Exercise 1 — Plan A WORKS (YES): draw the circle, add rays, colour it — each step has what it needs.

Exercise 2 — Plan B does NOT work (NO): eating is first, but you must pour the cereal then the milk before you can eat; eating should be last.

Exercise 3 — Plan C does NOT work (NO). A working order: climb the ladder, sit at the top, slide down (you must sit before you slide).

If students get stuck — and ways to adjust

Watch for: saying YES because the steps look familiar. Check each step has what it needs from earlier steps.

Make it easier: act out the broken plan so the problem is obvious.

Make it harder: ask them to explain exactly which step is too early and why.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students put step-by-step instructions in the right order to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a set of steps, change or reorder one, and explain what the change did.

You'll know it worked when

The child judges each plan correctly and rewrites the broken one into a working order.